

AI 2002

Artificial Intelligence

Lecture 1

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FAST NUCES CFD

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

INTELLIGENCE

The ability to learn and solve problems

ARTIFICIAL INTELLIGENCE:

The capacity given by human to machine to memorize and learn from experience, to think to create, to judge and make decision.

In other words,

1. The ability to solve problem.
2. The ability to act rationally.
3. The ability to act human.

What is Artificial Intelligence

“The **ability to learn or understand** things or to deal with new or difficult situations.” (**Merriam Webster**)

“The ability to learn, understand and make **judgments or have opinions** that are based on reason.” (**Cambridge Advance Learner’s Dictionary, 2006**)

“The ability to **acquire and apply knowledge** and skills.”(**Compact Oxford Dictionary, 2006**)

AI Application

Self-drive Cars:

- In 2005, a driverless robotic car named **STANLEY** sped through the rough terrain of the Mojave desert at 22 mph, finishing the 132-mile.
- Google self-driving car project **WAYMO**, stands for a *new way forward in mobility*.



Artificial Intelligence

AI can be defined with four different perspectives

- Systems that **think like humans**
- Systems that **act like humans**
- Systems that **think rationally**
- Systems that **act rationally**

Think like people



Think rationally



Act like people



Act rationally



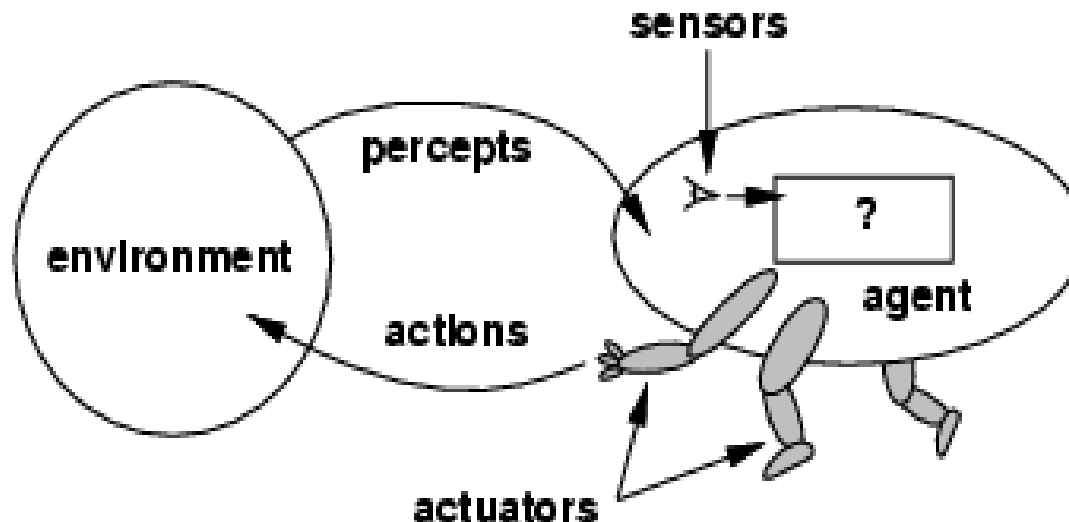
What is agent?

An Agent is an **entity** which

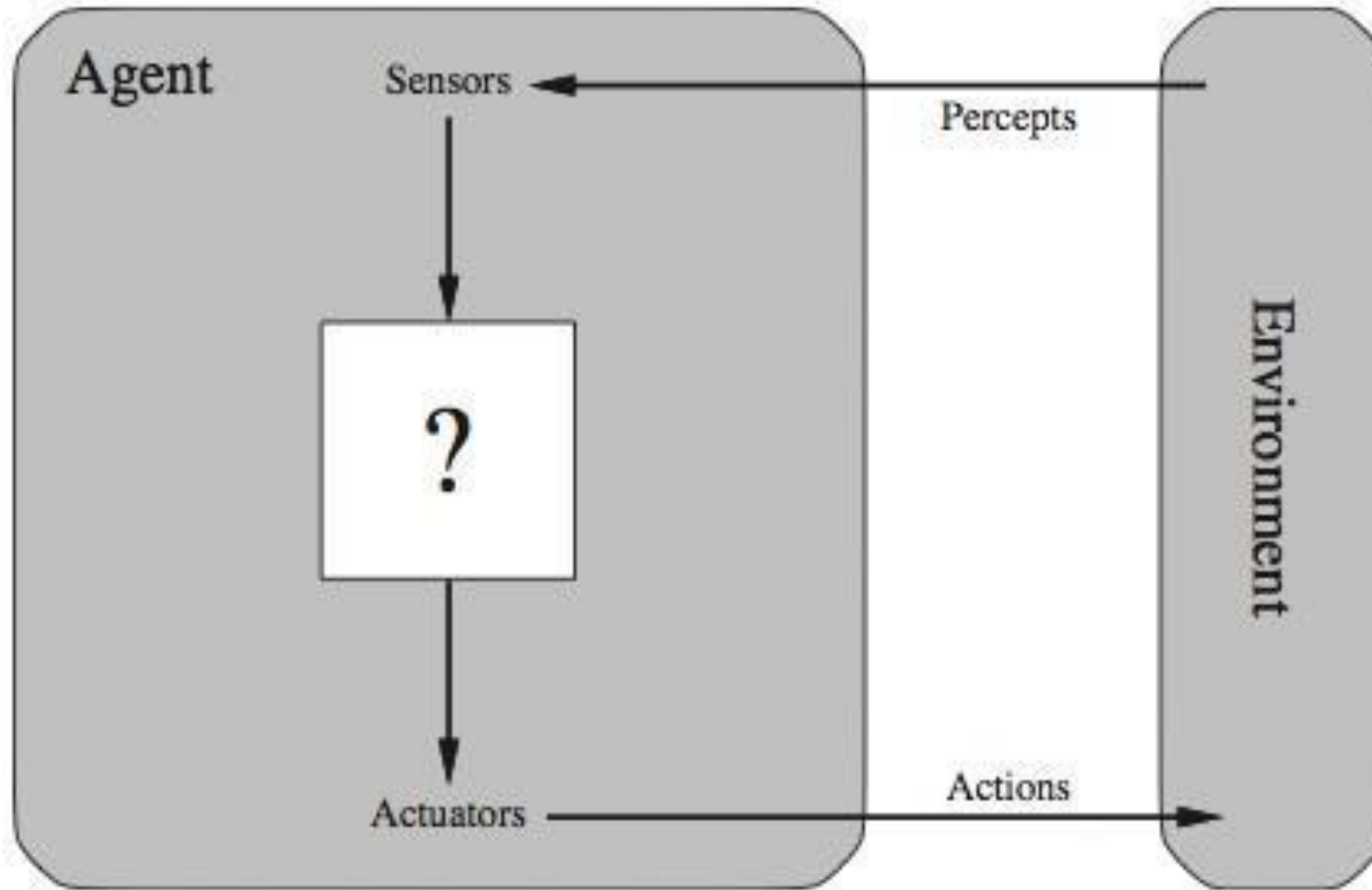
perceives its environment through **sensors** and

acts upon that environment through **actuators**

Assumption: Every agent can perceive its own actions (but not always the effects)



Agent with an Environment



Agents and environments

An agent is anything that;

- can be viewed as perceiving its environment through sensors and acting upon that environment through effectors (actuators).

A human agent has eyes, ears, and other organs for sensors hands, legs, mouth, and other body parts for effectors.

Agent

Abstractly, an agent is **a function which maps percept histories to actions:**

$$[f: P^* \rightarrow A]$$

Internally, the agent function will be implemented by **an agent program** which runs on the **physical architecture** to produce f

agent = **architecture** + **program**

Agent

Agent Examples:

Human agent:

Sensors --- eyes, ears, and other organs

Actuators --- hands, legs, mouth, and other body parts

Robotic agent:

Sensors --- cameras and infrared range finders

Actuators --- various motors for actuators

A software agent:

Sensors --- Keystrokes, file contents, received network packages

□ Actuators --- Displays on the screen, files, sent network packets

Example

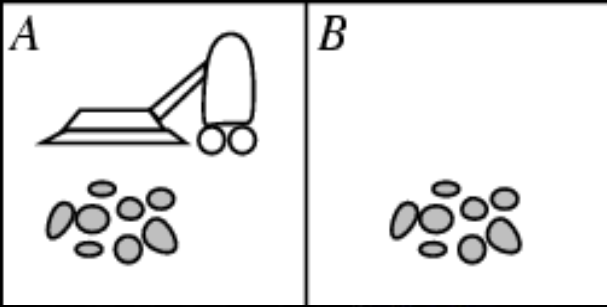
- Consider the case of an agent that is supposed to vacuum a dirty floor?

How Agents Should work?

- Tasks Parameters for Agents to perform accurately
- Environment Parameters for Agents to perform accurately

Agent Example

- Two locations: A and B
- **Precepts**
 - location
 - contents [A, Dirty]
- **Actions**
 - *Left*
 - *Right*
 - *Suck*
 - *NoOp*

A B	
	
Percept sequence	Actions
[A,Clean]	Right
[A, Dirty]	Suck
[B,Clean]	Left
[B,Dirty]	Suck
[A,Clean],[A,Clean]	Right
[A,Clean],[A,Dirty]	Suck
...	...
[A,Clean],[A.Clean],[A,Clean]	Right
[A,Clean],[A,Clean],[A, Dirty]	Suck

Rational Agent

- A rational agent is one that does the right thing.

How agent should behave?

Performance Measure: As a general rule, it is better to *design performance measures* according to what one actually wants in the environment.

Performance measure of a vacuum-cleaner agent could be *amount of dirt cleaned up, amount of time taken, amount of electricity consumed, amount of noise generated* etc.

Reward the agent for having a clean floor

Acting Rationally

- Rationality

What is rational at any given time?

- It depends on four things:

1. The **performance measure** that defines the criterion of success.
2. The **agent's prior knowledge of** the environment.
3. The **actions** that the agent can perform.
4. The agent's **percept sequence** to date

Tasks Parameters for Agents to perform accurately

- Major factors to evaluate the agents actions are;

Use PEAS to describe task

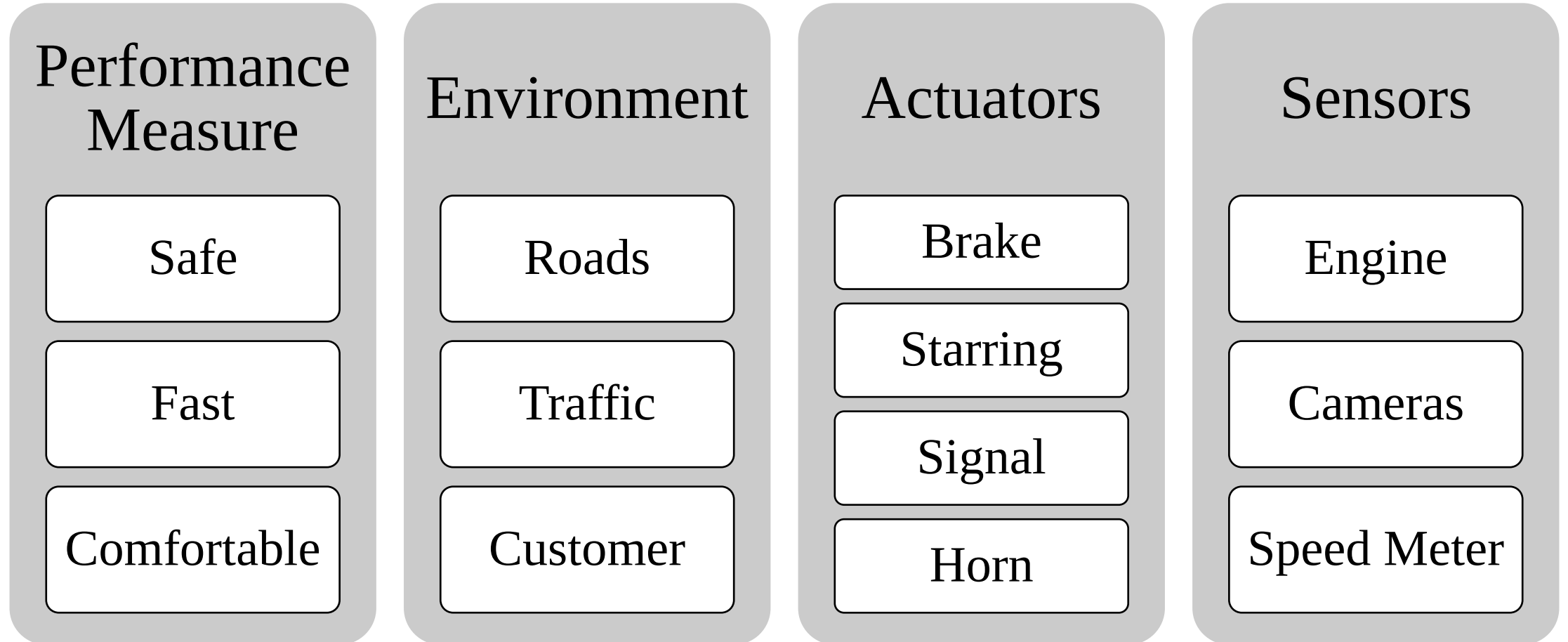
Performance measure

Environment

Actuators

Sensors

Example: Taxi Driver



Example: Medical Diagnosis

- **Performance measure:** Healthy patient, minimize costs etc.
- **Environment:** Patient, hospital, staff etc.
- **Actuators:** Screen display (questions, tests, diagnoses, treatments, referrals)
- **Sensors:** Keyboard (entry of symptoms, findings, patient's answers)

PEAS for satellite image analysis system

- **Performance measure:** correct image categorization
- **Environment:** downlink from the orbiting satellite
- **Actuators:** display categorization of scene
- **Sensors:** color pixel arrays

2. Environment Parameters for Agents to perform accurately

- i) Fully observable vs. partially observable
- ii) Deterministic vs. stochastic / strategic
- iii) Episodic vs. sequential
- iv) Static vs. dynamic
- v) Discrete vs. continuous
- vi) Single agent vs. multiagent

Fully Observed Vs Partially Observed

monitoring the specific tasks
(either Fully or partial).

- agent's sensory apparatus gives it
access to the complete/partial state
of the environment,

If we completely depend on sensors
then its called fully observed other
wise partially observed.

Example

Chess

Taxi Driver

Deterministic Vs Stochastic

- well-known knowledge of particular tasks. (either strategic or stochastic (change with condition)).
- If the next state of the environment is completely determined by the current state **then deterministic.**
 - Example: Rubik Cubic is acting as deterministic).
- If the next state of the environment is previously knowledge-based determined by the **current state then Strategic.**
 - Example: chess is acting as strategic).
- If the next state of the environment is uncertain situation determined by the **current state then Stochastic.**
 - Example: driving car is acting as stochastic).

Episodic vs. sequential

- series of separate parts (either sequential or episodic).
- the agent's experience is divided into "episodes."
- Each episode consists of the agent perceiving and then acting.

Lecture Delivering is
episodic

Playing chess is Sequential

Static Vs Dynamic

- Static environments
- An environment which no change are easy to deal with because the agent need not keep looking at the world.
 - Example: Medical diagnosis is acting as static).
- If the environment can change while an agent is deliberating (thinking), then we say the environment is dynamic.
 - Example :driving a car is acting as dynamic).

Discrete vs Continuous

Discrete

- There are a **fixed number of possible moves on each turn, then Discrete.**
 - Example: chess is acting as discrete

Continuous

- Agent **receives numerous range of continuous values, then its Continuous.**
 - Example: driving car is acting as continuous

Single vs Multi Agent

- **Single Agent**

- An agent operating by itself in an environment is a single agent or agent that responsible for performing a specific task.

- Example

- Crossword puzzle
- Virtual Assistant
- Chat Games

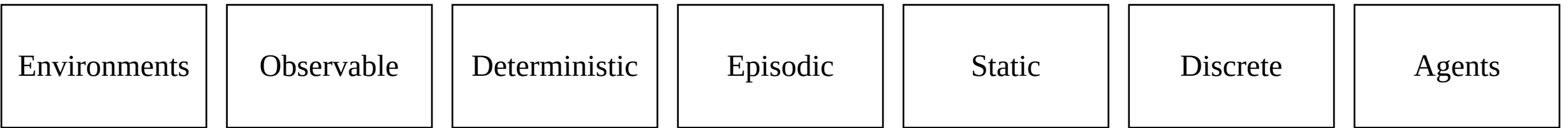
- **Multi Agent**

- A environment involving more than one agent is called multi agent environment.
- An agent playing chess with two players/

Environment Parameters for Agents to perform accurately

- Chess System
- Taxi Driver

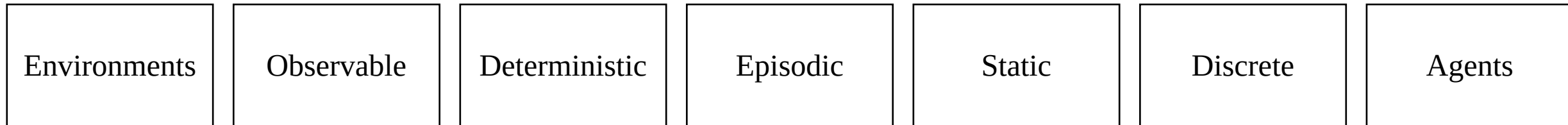
Environment Parameters for Agents to perform accurately (cont...)



Taxi Driver



Chess Game



Taxi Driver

Environments • Taxi Driver	Observable • Partial	Deterministic • Stochastic	Episodic • Sequential	Static • Dynamic	Discrete • Continuous	Agents • Single
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Chess Game

Environments • Playing with clock	Observable • Fully	Deterministic • Strategy	Episodic • Sequential	Static • Semi Dynamic	Discrete • Discrete	Agents • Multi
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Examples:

Environments	Observable	Deterministic	Episodic	Static	Discrete	Agents
Chess with a clock	Fully	Strategic	Sequential	Semi	Discrete	Multi
Taxi driving	Partial	Stochastic	Sequential	Dynamic	Continuous	Single
Medical diagnosis	Partial	Stochastic	Episodic	Static	Continuous	Single
Image analysis	Fully	Deterministic	Episodic	Semi	Discrete	Single
Robot part picking	Fully	Deterministic	Episodic	Semi	Discrete	Single
Interactive English tutor	Partial	Stochastic	Sequential	Dynamic	Discrete	Multi

Agent Types

- Five Types of Agents
 - Simple reflex agents
 - Reflex agents with state
 - Goal-based agents
 - Utility-based agents
 - Learning agent.